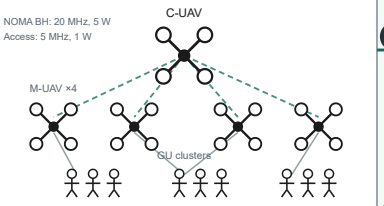
A. Environment

A1 Network model



A2 Dynamic sessions



20 → 130 active GUs

QoS 0.7 / 2 / 2.5 Mbit/s per class

session 45–120 s

random arrivals / departures

A3 Association

Rate-weighted K-means++ → cluster centers

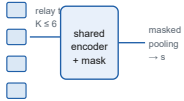
Hungarian matching → $x_{i,k}$, U_k



B. Branch-Credit Controller

B1 Relay-set encoder

$o_{k,n} \in \mathbb{R}^{14}$, k=1..K



B2 Motion actor

$\pi_M(a_M \mid s)$

stochastic, 3-D

B2 Twin critics

$Q_M^{\varphi 1}$, $Q_M^{\varphi 2}$

recursive soft target

B3 Power actor

$\pi_P(s)$

deterministic, K-D

B3 Same-slot utility

$u_P(s, a_P)$

no target critic

B4 Projection & training

flight limits, power softmax

$a_M=(v, \theta, \varphi)$, $a_P \rightarrow p_k^B$

replays D_M, D_P; SAC losses

Flow: ① state s_n ② actions a_M, a_P ③ capacities C_A, R_B ④ delivered D_i, C_n ⑤ QoS histories S_i, O_i

C. HA-QCS Scheduler

1 Eligibility & guard

$F_k = \{ i : C_i^A \geq R_i^{min} \}$

$a_i^{req} = R_i^{min} / C_i^A \leq 1$

rescue overdue $O_i \geq T_{guard}$

≤ 1 rescue per relay / slot

2 Cost-ordered completion

$\ell_i = R_i^{min} / C_i^A + R_i^{min} / R_k^B$

complete in ascending ℓ_i

if access + backhaul fit

3 Residual filling

history-weighted PF:

$w_i \propto 1 + O_i/5 + (1-S_i)/10$

leftover access / backhaul

per relay k, every slot: $\Sigma_i a_i^{req} \leq 1$, $\Sigma_i R_i^{min} \leq R_k^B$; output delivered rates D_i and completion C_n.

1 2 3 4 5